Learn from examples

AICE3002 / AICE6001

VLC = $\iiint$ Vision Learning Control $dV dL dC$

Basic Machine Learning I

Objectives, datasets, losses, and baselines

Jonathon Hare

Vision, Learning and Control
University of Southampton

What makes a problem supervised?

Inputs, targets, models, and predictions

From individual errors to empirical risk

Regression losses encode different assumptions

Classification requires scores and probabilities

Binary and multiclass cross-entropy

Baselines reveal whether learning is useful

Training, validation, and test data have different jobs

Dataset construction shapes the result

A complete supervised-learning workflow